ESS ENGINEERING STEEL SOLUTIONS

301 STAINLESS STEEL

Type 301 is an austenitic chromium-nickel stainless steel. This alloy is nonmagnetic in the annealed condition, but becomes magnetic when cold worked. 301 has a nominal composition of 17 percent chromium and 7 percent nickel. Type 301 is available annealed and in a variety of temper-rolled conditions.

High strength and excellent corrosion resistance make this versatile grade suitable for automobile trim, hose clamps, springs etc.

Resistance to oxidation

Type 301 possesses good resistance to oxidation at temperatures up to 1550°F (840°C). At 1600°F (871°C), Type 301 exhibits an oxidation weight gain of 10mg/cm2 in 1,000 hours. Therefore, this stainless steel is not suggested for use at 1600°F.

Chemical composition

Represented by ASTM A240 and A666

Element	Percent by Weight
Carbon	0.15 maximum
Manganese	2.00 maximum
Phosphorus	0.045 maximum
Sulfur	0.030 maximum
Silicon	0.75 maximum
Chromium	16.00-18.00
Nickel	6.00-8.00
Nitrogen	0.10 maximum
Iron	Balance

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Hardness

Typical hardness values for annealed and cold-rolled Type 301 are given in the following table:

Temper	Brinell Hardness	Rockwell Hardness
Annealed	165	B86
1/4 Hard	255	B25
1/2 Hard	297	B32
3/4 Hard	342	B37
Full Hard	383	B41

Mechanical properties

Condition	UTS ksi (MPa) min*	0.2% YS ksi (MPa) min*	Elongation % in 2" (50.8 mm)
Annealed	120 (827)	45 (310)	60
1/4 Hard	125 (862)	75 (517)	25
1/2 Hard	150 (1034)	110 (758)	18
3/4 Hard	175 (1207)	135 (931)	12
Full Hard	185 (1276)	140 (965)	9

* Minimum - standard practice is to produce to either minimum tensile strength, minimum yield strength or minimum hardness, but not to combinations of these properties.

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Physical properties

Density, lbs./in.3 (g/cm3)	0.285 (7.88)
Electrical Resistivity, µΩ•in. (µΩ•cm)	27.4 (69.5
Thermal Conductivity, BTU/hr./ft./°F (W/m/K)	
212 °F (100 °C)	9.4 (16.2)
932 °F (500 °C)	12.4 (21.4)
Modulus of Elasticity, ksi (MPa)	
in tension	28.0 x 10^3 (193 x 10^3)
Shear modulus in torsion	11.2 x 10^3 (178 x 10^3)
Poissons ratio	0.24
Magnetic Permeability, (H/m at 200 Oersteds)	Annealed 1.02 max
Specific Heat, BTU/lbs./°F (kJ/kg/K)	
32–212 °F (0–100°C)	0.12 (0.50)
Melting Range, °F (°C)	2250 – 2590 (1399 – 1421)

Heat Treatments

Type 301 is non-hardenable by heat treatment. Annealing: Heat to 1900 – 2050 °F (1038 – 1121 °C), then water quench. Stress Relief Annealing: Heat to 500 – 900 °F (260 – 482 °C), then air cool.

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Facility information

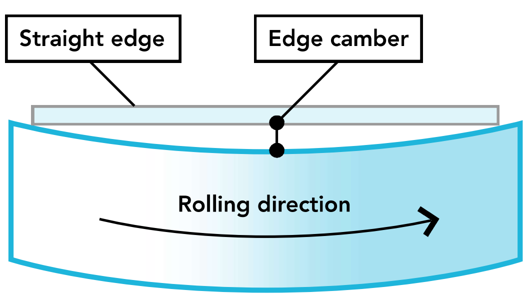
- 50 000 square foot operating facility
- 18 feet under the crane rail
- Main coil bay 280 feet in length
- Crane capacity 15 Ton

Delivery format

Thickness	0.012″ – 0.40″
Thickness tolerance	Normal +/- 10 %
	Improved +/- 5 %
Minimum slit width	0.6″
Width tolerance	+/- 0.005″
Max weight	Payoff 35 000 lbs
	Recoil 35 000 lbs
Max coil OD	Up to 72″
Inner diameter	16″ and 20″
Burrs	Less than 10 % of strip thickness
Papper interleave / plastic	With or without
Cardboard core	With or without
Delivery form	Pancake coils
Flatness	Tension leveled material
* **Camber**	Max 0.5″ / 8 feet
Surface	2B or 2H

*

Edge camber = the greatest deviation of a side edge from a straight line, the measurement being taken on the concave side with a straight edge



Edge camber